

Transport-Coupled Bayesian Flows for Molecular Graph Generation

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Abstract

Molecular graph generation (MGG) is essentially a multi-class generative task, aimed at predicting categories of atoms and bonds under strict chemical and structural constraints. However, many prevailing diffusion paradigms learn to regress numerical embeddings and rely on a hard discretization rule during sampling to recover discrete labels. This introduces a fundamental discrepancy between training and sampling. While models are trained for point-wise numerical fidelity, the sampling process fundamentally relies on crossing categorical decision boundaries. This discrepancy forces the model to expend efforts on intra-class variations that become irrelevant after discretization, ultimately compromising diversity, structural statistics, and generalization performance. Therefore, we propose TopBF, a unified framework that (i) performs MGG directly in continuous parameter distributions, (ii) learns graph-topological understanding through a Quasi-Wasserstein optimal-transport coupling under geodesic costs, and (iii) supports controllable, property-conditioned generation during sampling without retraining the base model. TopBF innovatively employs cumulative distribution function (CDF) to compute category probabilities induced by the Gaussian channel, thereby unifying the training objective with the sampling discretization operation. Experiments on QM9 and ZINC250k demonstrate superior structural fidelity and efficient generation with improved performance. The formal theoretical proofs, experimental details, and supplementary experiments are provided in the appendices.

1. Introduction

Molecular graph generation (MGG) is a critical yet challenging endeavor in modern computational drug discovery (Li et al. 2025a), facilitating the advancement of downstream tasks such as molecular optimization (Zhu et al. 2023; Xiong et al. 2026) and drug–target binding affinity prediction (Wu et al. 2024; Li et al. 2025b). Early computational approaches utilizing deep learning attempted to generate molecular graphs in a one-shot manner (Zang and Wang 2020; Kong et al. 2022). However, these methods often struggle to model complex structures effectively. In recent years, diffusion models have demonstrated remarkable performance across various generation tasks (Gong et al. 2024; Chen et al. 2025). These models are generally divided into discrete and continuous types and have been progressively integrated into molecular graph generation.

Naturally suited for discrete graph structures, discrete diffusion models typically define a forward Markov process that corrupts the graph structure through discrete edits. Conversely, the reverse process learns to restore the graph by framing generation as an iterative classification task (Vignac et al. 2023; Xu et al. 2024). However, these discrete diffusion models sacrifice the smoother

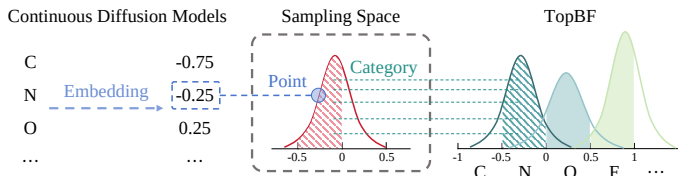


Figure 1 Continuous diffusion models are agnostic of the discretization process and only pursue numerical fitting during training, while TopBF directly calculates the probability of each category.

optimization and dynamic generation afforded by continuous representations. Continuous diffusion models have also demonstrated success in this domain (Jo et al. 2022; Luo et al. 2023). They operate on a continuous representation of a graph, involving a forward noising process that incrementally perturbs this representation with continuous noise while training a core network to reverse this corruption. The primary objective of continuous diffusion models is to predict embedded representations in continuous space and train a neural network to regress those values under injected noise. However,

the final output must be discretized by rounding, which is not a benign post-processing step. As shown in Figure 1, it is evident that throughout the entire training phase, these models remain agnostic regarding the discretization rule. The resulting mismatch between training and sampling can lead to inefficient learning and brittle generation. In particular, point-wise regression errors near category boundaries may be penalized mildly during training but induce catastrophic category flips at inference, which is especially problematic for generation.

Recently, Bayesian Flow Networks (BFNs) (Graves et al. 2023; Song et al. 2024) provide an alternative paradigm for discrete generation by evolving distribution parameters through a continuous-time Bayesian updating process. Instead of directly predicting discrete values, BFNs propagate a parameter state that represents uncertainty and update it via pseudo-observations, yielding a continuous and differentiable learning process for discrete data. However, BFNs are not topology-aware for molecular graphs, since they adopt independent Gaussian distributions between variables to retain closed-form Bayesian updates. As a result, BFNs are mainly rewarded for getting each position right in isolation, rather than for capturing how labels should cohere with the overall graph structure.

To address the aforementioned issues, we introduce TopBF, a molecular graph generative BFN that performs generation in continuous distribution-parameter space and uses a CDF-based decoding to compute category probabilities from continuous parameters, thereby aligning the training objective with the same discretization rule used during sampling. Furthermore, we make TopBF topology-aware by introducing a Quasi-Wasserstein transport coupling defined on graph geodesic costs, which compares predicted categorical mass on atoms and bonds with the ground-truth structure. This geometry-respecting discrepancy encourages local, structure-consistent adjustments and discourages topology-inconsistent long-range rearrangements. TopBF also achieves conditional generation via property-error gradient (PEG) guidance. Notably, TopBF achieves superior performance with the minimum sampling steps, accelerating molecular graph generation in practice. We summarize our key contributions as follows:

- We observe the mismatch defect in continuous diffusion methods and propose TopBF that unifies continuous-parameter training with discrete sampling for MGG.
- We inject topology awareness into TopBF training via a Quasi-Wasserstein transport coupling with graph-geodesic ground costs by aligning predicted categorical mass with the molecular structure in a geometry-consistent manner.
- TopBF supports training-free and property-conditioned generation via guided sampling while requiring the minimum sampling steps, achieving both effective control and efficient MGG.

2. Related Work

Deep generative models for MGG are typically categorized into autoregressive and one-shot models. The former generate graphs incrementally, adding atoms and bonds in an autoregressive manner (You et al. 2018a,b; Jin et al. 2018, 2020). The latter generate the entire graph at once (Simonovsky and Komodakis 2018; Shi et al. 2020; Luo et al. 2021; Martinkus et al. 2022).

Diffusion models have been adapted for molecular graph generation (Liu et al. 2023; Qiu et al. 2023; Li et al. 2024). Some methods define

a diffusion process directly on the discrete state space (Vignac et al. 2023; Xu et al. 2024), which can be complex to accomplish. Others map discrete attributes to a continuous space and apply standard Gaussian diffusion (Huang et al. 2022; Lee et al. 2023).

BFNs have arisen recently to perform generation in continuous parameter distribution (Graves et al. 2023; Song et al. 2024; Qu et al. 2024). The process starts with a simple prior distribution and iteratively refines its parameters via Bayesian inference to calculate the posterior from observed samples.

3. Preliminaries

3.1. Problem Definition

A molecular graph G is defined as (X, A) , where $X \in \{1, \dots, K_X\}^D$ represents a matrix of D atom features derived from K_X classes, and $A \in \{1, \dots, K_A\}^{D \times D}$ denotes an adjacency matrix encapsulating bond features from K_A classes. Here D is the maximum number of atoms in the considered molecules and let $N \leq D$ be the number of valid atoms.

Given a dataset $\mathcal{D} = \{G_i\}_{i=1}^{|\mathcal{D}|}$ sampled from an unknown data distribution $p_{\text{data}}(G)$, the goal of molecular graph generation is to learn a generative model $p_{\theta}(G)$ that can efficiently sample valid, diverse molecular graphs whose distribution matches p_{data} in terms of structural and chemical statistics. For the discrete nature of molecular graphs, this task can be seen as a multi-class generative task, where we need to assign specific atomic or chemical bond types at each position subject to chemical validity constraints. In addition, the model may be guided towards desired properties $\mathbf{z}$ to achieve conditional molecular graph generation as a flexible extension.

3.2. Bayesian Flow Networks

Bayesian Flow Networks can be conceptualized as a data transmission protocol between a sender and a receiver. Given data $\mathbf{x} = (x^{(1)}, \dots, x^{(D)})$, let $\boldsymbol{\theta} = (\theta^{(1)}, \dots, \theta^{(D)})$ be the parameters of a factorized input distribution: $p_I(\mathbf{x} | \boldsymbol{\theta}) = \prod_{d=1}^D p_I(x^{(d)} | \theta^{(d)})$. At each time $t \in [0, 1]$, the sender corrupts $\mathbf{x}$ by a Gaussian channel α yielding a sender distribution: $p_S(\mathbf{y} | \mathbf{x}; \alpha) = \prod_{d=1}^D p_S(y^{(d)} | x^{(d)}; \alpha)$. During the data transmission process, the input parameters $\boldsymbol{\theta}$ are input into a neural network Ψ , whose output is then parameterized as an output distribution: $p_O(\mathbf{x} | \boldsymbol{\theta}, t) = \prod_{d=1}^D p_O(x^{(d)} | \Psi(\boldsymbol{\theta}, t))$. Since the receiver knows the form of the sender distribution but does not know $\mathbf{x}$, it integrates all possible $\mathbf{x}'$ and sender distributions, weighted by the output distribution, and obtains the receiver distribution: $p_R(\mathbf{y} | \boldsymbol{\theta}; t, \alpha) = \mathbb{E}_{p_O(\mathbf{x}' | \boldsymbol{\theta}; t)} p_S(\mathbf{y} | \mathbf{x}'; \alpha)$. Subsequently, by applying Bayesian inference, Bayesian update function $h(\cdot)$ is derived to repeatedly update the parameters: $\boldsymbol{\theta}' = h(\boldsymbol{\theta}, \mathbf{y}, \alpha)$ along a noise schedule, defined as a continuous-time Bayesian flow $p_F(\boldsymbol{\theta} | \mathbf{x}; t)$. The Bayesian update distribution is then defined by marginalizing out $\mathbf{y}$: $p_U(\boldsymbol{\theta}' | \boldsymbol{\theta}, \mathbf{x}; \alpha) = \mathbb{E}_{p_S(\mathbf{y} | \mathbf{x}; \alpha)} \delta(\boldsymbol{\theta}' - h(\boldsymbol{\theta}, \mathbf{y}, \alpha))$, where $\delta(\cdot)$ is the Dirac delta distribution. Now define the accuracy schedule $\beta(t)$ as $\beta(t) = \int_{t'=0}^t \alpha(t') dt'$, and the Bayesian flow distribution is the marginal distribution over input parameters at time t : $p_F(\boldsymbol{\theta} | \mathbf{x}; t) = p_U(\boldsymbol{\theta} | \boldsymbol{\theta}_0; \mathbf{x}; \beta(t))$.

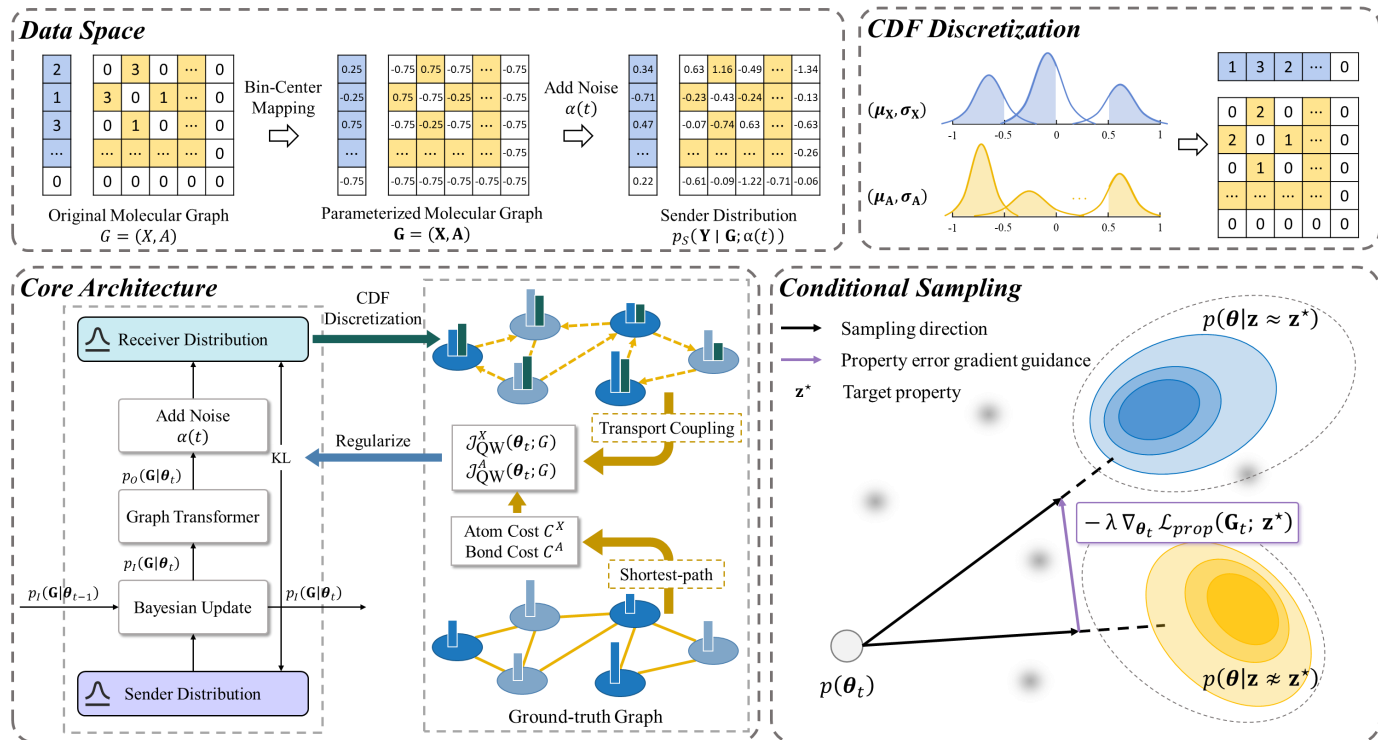


Figure 2 Overall framework of TopBF. Data Space: The procedure of parameterizing and adding noise to the molecular graph for the sender distribution. CDF Discretization: Obtain atom and bond categories from the output of Ψ through CDF. Core Architecture: The structure of a BFN module at time t . And a QW transport coupling regularizes the training procedure. Conditional Sampling: PEG guidance for conditional generation.

3.3. Optimal Transport and Quasi-Wasserstein Coupling on Graphs

Optimal transport (OT) provides a geometry-aware way to compare two distributions by measuring the minimum cost required to move probability mass from one to the other (Gabriel and Marco 2019). Consider a molecular graph $G = (X, A)$ with N atoms, and let $C \in \mathbb{R}_+^{N \times N}$ be the geodesic ground-cost matrix induced by the shortest-path metric on G , i.e., $C_{ij} = \text{dist}_G(i, j)$. Given two probability distributions $\mathbf{a}, \mathbf{b} \in \Delta^{N-1}$ over atom indices, the entropically regularized OT cost is $\mathcal{W}_\varepsilon(\mathbf{a}, \mathbf{b}; C) = \min_{P \in \Pi(\mathbf{a}, \mathbf{b})} \langle P, C \rangle + \varepsilon \sum_{i,j} P_{ij} (\log P_{ij} - 1)$, where $\Pi(\mathbf{a}, \mathbf{b}) = \{P \geq 0 \mid P\mathbf{1} = \mathbf{a}, P^\top \mathbf{1} = \mathbf{b}\}$. In practice, we solve the entropy-regularized OT and obtain the coupling P via Sinkhorn iterations (Cuturi 2013).

Quasi-Wasserstein (QW) extends this idea from comparing single scalar signals on atoms to comparing multi-class label fields on graphs. Specifically, for a categorical field with K classes, we view the category probability map $P(\cdot, k) \in [0, 1]^N$ as a mass distribution over atom indices for each class k , and compare prediction $P(\cdot, k)$ and target $Q(\cdot, k)$ by transporting mass within each class under the same geodesic cost C . The QW discrepancy then aggregates these per-class transport costs: $\mathcal{D}_{QW}(P, Q; C) = \frac{1}{K} \sum_{k=1}^K \mathcal{W}_\varepsilon(\mathbf{a}_k, \mathbf{b}_k; C)$, $\mathbf{a}_k = \text{Norm}(P(\cdot, k))$, $\mathbf{b}_k = \text{Norm}(Q(\cdot, k))$, where $\text{Norm}(\cdot)$ denotes normalization to the simplex on valid atoms. In contrast to position-wise divergences, $\mathcal{D}_{QW}$ penalizes errors according to how far probability mass must move along the graph, thereby encouraging local and topology-consistent corrections while discouraging long-range and structure-inconsistent reallocations.

4. Materials and Methods

In this section, we describe how our MGG framework, Transport-Coupled Bayesian Flows (TopBF), generates molecular graphs via Bayesian flows over continuous distribution parameters, as illustrated in Figure 2. Pseudocodes for training and sampling are provided in Appendix 8.

4.1. Datasets and Metrics

We validate our approach on the QM9 (Ramakrishnan et al. 2014) and ZINC250k (Irwin et al. 2012) benchmarks, following the previous experimental setup (Jo et al. 2022). QM9 contains approximately 133,885 molecules with up to 9 heavy atoms of 4 types, while ZINC250k comprises roughly 249,455 molecules with up to 38 heavy atoms of 9 types, both covering three standard bond types. Adhering to the standard pipeline (Shi et al. 2020; Luo et al. 2021; Jo et al. 2022), molecules undergo kekulization via RDKit, with explicit hydrogens excluded to focus on the heavy-atom graph representation.

We evaluate the experiments following previous evaluation setting (Jo et al. 2022): Validity without correction and uniqueness on 10,000 generated molecules. Fréchet ChemNet Distance (FCD) (Preuer et al. 2018) measures the distance between training and generated sets using the activations from the penultimate layer of the ChemNet. Additionally, Neighborhood Subgraph Pairwise Distance Kernel (NSPDK) maximum mean discrepancy (MMD) (Costa and De Grave 2010) evaluates the MMD between the generated molecules and test molecules which takes into account both the atom and bond features for evaluation.

4.2. Parameterization of Molecular Graphs

Since directly manipulating discrete data often leads to more complex operations to guarantee differentiability, a foundational step of our method involves mapping the discrete atom and bond features into continuous spaces. For a feature class $k \in \{1, \dots, K\}$, we define a deterministic mapping to a continuous value $x_k \in [-1, 1]$:

$$k_c = \frac{2k-1}{K} - 1, \quad k_l = k_c - \frac{1}{K}, \quad k_r = k_c + \frac{1}{K}, \quad (1)$$

where k_l , k_c and k_r represent the respective vectors for left boundaries, centers and right boundaries of the classes. Importantly, we assign the continuous value $x_k := k_c$ to the k -th class. This mapping transforms the ground-truth graph G into continuous representation $\mathbf{G} = (\mathbf{X}, \mathbf{A})$. This continuous representation serves as the target signal for the generative flow.

4.3. Bayesian Flow Over Distribution Parameters

4.3.1. Input distribution in parameter space.

Instead of modeling G directly as diffusion models, TopBF operate on parameters of a simple input distribution over continuous $\mathbf{G}$. For each of atoms and bonds we maintain a Gaussian input distribution:

$$p_I(\mathbf{X} | \boldsymbol{\theta}_X) = \mathcal{N}(\mathbf{X} | \boldsymbol{\mu}_X, \rho_X^{-1} \mathbf{I}), \quad (2)$$

$$p_I(\mathbf{A} | \boldsymbol{\theta}_A) = \mathcal{N}(\mathbf{A} | \boldsymbol{\mu}_A, \rho_A^{-1} \mathbf{I}), \quad (3)$$

where $\boldsymbol{\theta} \stackrel{def}{=} (\boldsymbol{\mu}, \rho)$ consists of a mean vector $\boldsymbol{\mu} \in \mathbb{R}^D$ and a scalar precision $\rho > 0$, and $\mathbf{I}$ is the $D \times D$ identity matrix. The process initiates with a standard normal prior $p_I(\cdot | \boldsymbol{\theta}_0) = \mathcal{N}(\cdot | \mathbf{0}, \mathbf{I})$ at time $t = 0$. For brevity, we will use the integrated form in our formulas: $p_I(\mathbf{G} | \boldsymbol{\theta}) = \mathcal{N}(\mathbf{G} | \boldsymbol{\mu}, \rho^{-1} \mathbf{I})$, except in special circumstances.

4.3.2. Sender distribution: adding noise to the data.

Given continuous $\mathbf{G} = (\mathbf{X}, \mathbf{A})$, the sender distribution plays a role forwarding noising data. For each time-dependent accuracy parameter $\alpha(t)$, the sender draws a noisy message:

$$p_S(\mathbf{Y} | \mathbf{G}; \alpha(t)) = \mathcal{N}(\mathbf{Y} | \mathbf{G}, \alpha(t)^{-1} \mathbf{I}), \quad (4)$$

where $\mathbf{Y} = (\mathbf{Y}^{(1)}, \dots, \mathbf{Y}^{(D)})$ is a noisy observation of the true graph, whose noise level increases when $\alpha(t)$ is small.

4.3.3. Bayesian update and Bayesian flow distribution.

BFNs define a Bayesian update that combines the current input distribution $p_I(\mathbf{G} | \boldsymbol{\theta})$ with a noisy observation $\mathbf{Y}$ from the sender, yielding a posterior over $\mathbf{G}$ that can again be written as a Gaussian in closed form. Since both p_I and p_S are Gaussians with diagonal covariances, the Bayesian update for each of atoms and bonds reduces to (Murphy 2007):

$$\begin{cases} \rho_i = \rho_{i-1} + \alpha, \\ \boldsymbol{\mu}_i = \frac{\rho_{i-1} \boldsymbol{\mu}_{i-1} + \alpha \mathbf{Y}}{\rho_{i-1} + \alpha}. \end{cases} \quad (5)$$

Repeating this update along a schedule of accuracies induces a stochastic trajectory $\{\boldsymbol{\theta}_t\}_{t \in [0,1]}$ in parameter space that converges towards the true graph $\mathbf{G}$ as $t \rightarrow 1$.

Rather than explicitly simulating all intermediate steps, there exists a Bayesian flow distribution $p_F(\boldsymbol{\theta} | \mathbf{G}; t)$ describing the

distribution of input parameters at time t after aggregating all past Bayesian updates, as elaborated in Sec. 3.2.

Conceptually, the dynamics are defined on distribution parameters $(\boldsymbol{\mu}, \rho)$ rather than on data itself. This is the key distinction of TopBF from standard diffusion models.

4.4. Output and Receiver Distributions for Molecular Graphs

The core neural network in TopBF takes as input the current parameter state $\boldsymbol{\theta}_t$ and time t , and outputs a prediction of the Gaussian noise used to generate the input mean. We denote the network by:

$$\boldsymbol{\epsilon} = \Psi(\boldsymbol{\theta}_t, \mathbf{c}, t), \quad (6)$$

where $\mathbf{c}$ represents any optional conditioning. The output $\boldsymbol{\epsilon}$ is split into two D -dimensional vectors $(\boldsymbol{\mu}_\epsilon, \ln \sigma_\epsilon)$, which are then transformed into the mean and standard deviation $(\hat{\boldsymbol{\mu}}_\mathbf{G}, \hat{\sigma}_\mathbf{G})$ of a Gaussian over the continuous data (formal derivation of the transformation in Appendix 1):

$$\begin{aligned} \hat{\boldsymbol{\mu}}_\mathbf{G} &= \begin{cases} \mathbf{0} & \text{if } t < t_{\min}, \\ \frac{\boldsymbol{\mu}}{\gamma(t)} - \sqrt{\frac{1-\gamma(t)}{\gamma(t)}} \boldsymbol{\mu}_\epsilon & \text{otherwise,} \end{cases} \\ \hat{\sigma}_\mathbf{G} &= \begin{cases} \mathbf{1} & \text{if } t < t_{\min}, \\ \sqrt{\frac{1-\gamma(t)}{\gamma(t)}} \exp(\ln \sigma_\epsilon) & \text{otherwise,} \end{cases} \end{aligned} \quad (7)$$

where $\gamma(t) = \beta(t)/(1 + \beta(t))$ is determined by the accuracy schedule $\beta(t)$ of BFNs, and $t_{\min}$ is a small positive constant to avoid numerical issues at $t = 0$.

4.4.1. Truncated Gaussian CDF and per-class probabilities.

While Eq. (7) defines a continuous Gaussian distribution over $\mathbf{G}$, our ultimate goal is to make discrete predictions over K classes. To bridge this gap, TopBF uses a truncated Gaussian cumulative distribution function (CDF) to compute the probability mass assigned to each discretization class.

For each dimension $d \in \{1, \dots, D\}$ we define the Gaussian CDF:

$$F(x | \mu_x^{(d)}, \sigma_x^{(d)}) = \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{x - \mu_x^{(d)}}{\sqrt{2} \sigma_x^{(d)}} \right) \right], \quad (8)$$

where $\operatorname{erf}(\cdot)$ is the error function defined as $\operatorname{erf}(u) = \frac{2}{\sqrt{\pi}} \int_0^u e^{-t^2} dt$. Here truncate the support to the interval $[-1, 1]$ to align with the embedding range:

$$\mathcal{F}(x | \mu_x^{(d)}, \sigma_x^{(d)}) = \begin{cases} 0 & \text{if } x \leq -1, \\ 1 & \text{if } x \geq 1, \\ F(x | \mu_x^{(d)}, \sigma_x^{(d)}) & \text{otherwise,} \end{cases} \quad (9)$$

Using the class boundaries (k_l, k_r) from Eq. (1), the probability of assigning dimension d to category $k \in \{1, \dots, K\}^D$ is determined by

the integral of the truncated Gaussian over the corresponding class:

$$p_O^{(d)}(k | \theta_t) = \mathcal{F}(k_r | \mu_x^{(d)}, \sigma_x^{(d)}) - \mathcal{F}(k_l | \mu_x^{(d)}, \sigma_x^{(d)}). \quad (10)$$

To achieve closed-form Bayesian update, the output distribution over the $\mathbf{G}$ is independent across dimensions:

$$p_O(\mathbf{G} | \theta_t) = \prod_{d=1}^D p_O^{(d)}(k^{(G^{(d)})} | \theta_t), \quad (11)$$

where $k(x^{(d)})$ denotes the index of the class occupied by $x^{(d)}$. Similarly, we define the expected class-center representation under the current output distribution as:

$$\begin{aligned} \mathbb{E}[P(\theta, t)] &= \hat{\mathbf{k}}(\theta, t) \\ &\stackrel{def}{=} \left(\sum_{k=1}^K p^{(1)}(k | \theta, t) k_c, \dots, \sum_{k=1}^K p^{(D)}(k | \theta, t) k_c \right). \end{aligned} \quad (12)$$

4.4.2. Receiver distribution as a mixture of Gaussians.

The receiver distribution p_R inverts the sender process in parameter space by describing the distribution over noisy messages $\mathbf{Y}$ implied by the current parameter state θ_t . Consequently, substituting Eqs. (4) and (10) into the BFN formulation yields a mixture of Gaussians:

$$\begin{aligned} p_R(\mathbf{Y} | \theta; t, \alpha(t)) &= \mathbb{E}_{p_O(\mathbf{G} | \theta_t)} p_S(\mathbf{Y} | \mathbf{G}; \alpha(t)), \\ &= \prod_{d=1}^D \sum_{k=1}^K p_O^{(d)}(k | \theta_t) \mathcal{N}(\mathbf{Y}^{(d)} | k_c, \alpha(t)^{-1}). \end{aligned} \quad (13)$$

The receiver plays the role of the reverse channel in the flow. During training, we encourage the receiver distribution to match the sender distribution applied to $\mathbf{G}$, while during sampling it serves to generate pseudo-observations that drive the Bayesian updates in Eq. (5). Notably, TopBF applies the above construction independently to atoms and bonds by using $K = K_X$ and $K = K_A$, respectively.

4.5. Topology-aware Optimal Transport Matching

Although the position-wise factorization in Eq. (11) enables closed-form Bayesian updates, it also yields largely local and index-wise supervision that does not explicitly encode graph topology. However, categorical errors are not equally severe, since swapping mass within a local neighborhood is structurally less disruptive than relocating it across distant substructures (Frogner et al. 2015; Cheng and Xu 2024). To bridge this gap, we introduce an OT coupling that regularizes predictions under molecule’s intrinsic geometry.

4.5.1. Graph-induced Ground Cost.

We define the atom-level ground cost by the shortest-path metric:

$$C_{ij}^X = \text{dist}_G(i, j), \quad i, j \in \{1, \dots, N\}. \quad (14)$$

For bonds, we define an bond-level cost C^A using an endpoint-based proxy consistent with our implementation, e.g., for two undirected bonds $e = (i, j)$ and $e' = (u, v)$,

$$C_{e,e'}^A = \min \{ \text{dist}_G(i, u), \text{dist}_G(i, v), \text{dist}_G(j, u), \text{dist}_G(j, v) \}. \quad (15)$$

Theorem 1 (Endpoint proxy matches line-graph geodesics) Let $\mathcal{L}(G)$ be the line graph of G whose vertices correspond to bonds and edges connect bonds that share an endpoint. For any two distinct undirected bonds e, e' ,

$$\text{dist}_{\mathcal{L}(G)}(e, e') = 1 + C_{e,e'}^A.$$

Thus, C^A differs from the exact line-graph geodesic only by an additive constant and induces the same optimal OT coupling (proof in Appendix 3).

4.5.2. Optimal Transport over Categorical Mass.

We make topology awareness a first-class objective by minimizing the transport work needed to align predicted categorical mass with the ground truth (Peyré et al. 2016). At time t , a sample $\theta_t \sim p_F(\theta | \mathbf{G}; t)$ yields categorical probabilities from Eq. (11), with targets Q_X, Q_A . We form class-wise mass fields by normalizing $P_X(\cdot, k)$ and $P_A(\cdot, k)$ on valid indices and only aggregate over classes that appear in the true graph G : $\mathcal{K}_X^+(G) = \{k | \exists i \text{ s.t. } X_i = k\}$ and $\mathcal{K}_A^+(G) = \{k | \exists e \text{ s.t. } A_e = k\}$. Thus we define the QW transport costs

$$\mathcal{J}_{\text{QW}}^X(\theta_t; G) = \frac{1}{|\mathcal{K}_X^+(G)|} \sum_{k \in \mathcal{K}_X^+(G)} \mathcal{W}_\varepsilon(\mathbf{a}_k^X, \mathbf{b}_k^X; C^X), \quad (16)$$

$$\mathcal{J}_{\text{QW}}^A(\theta_t; G) = \frac{1}{|\mathcal{K}_A^+(G)|} \sum_{k \in \mathcal{K}_A^+(G)} \mathcal{W}_\varepsilon(\mathbf{a}_k^A, \mathbf{b}_k^A; C^A), \quad (17)$$

where $\mathcal{W}_\varepsilon$ is the entropic OT cost in Sec. 3.3; for bonds we exclude the invalid bond class from $\mathcal{K}_A^+(G)$ to avoid trivial domination. The overall training objective then minimizes the expected transport work induced by the receiver predictions,

Theorem 2 (Hard-label limit of class-wise transport) Fix a class k and suppose $\mathbf{a}_k = \frac{1}{r} \sum_{\ell=1}^r \mathbf{e}_{i_\ell}$ and $\mathbf{b}_k = \frac{1}{r} \sum_{\ell=1}^r \mathbf{e}_{j_\ell}$ are uniform sums of r Diracs. Then

$$\mathcal{W}_0(\mathbf{a}_k, \mathbf{b}_k; C) = \frac{1}{r} \min_{\pi \in \mathfrak{S}_r} \sum_{\ell=1}^r C_{i_\ell, j_{\pi(\ell)}}.$$

Proof is in Appendix 4.2.

4.6. Training Objective

The BFN loss can be interpreted either as the number of nats required to transmit the data through the Bayesian flow, or equivalently as the negative variational lower bound over message sequences. Concretely, we sample $t \sim \mathcal{U}(0, 1)$ and $\theta_t \sim p_F(\theta | \mathbf{G}; t)$, and minimize the expected divergence between the sender distribution applied to the ground-truth discretized graph and the receiver distribution implied by θ_t :

$$\mathcal{L}_G = \mathbb{E}_{t, \theta_t} \left[\text{KL} \left(p_S(\mathbf{Y} | \mathbf{G}; \alpha(t)) \| p_R(\mathbf{Y} | \theta_t; t, \alpha(t)) \right) \right]. \quad (18)$$

In TopBF, p_S and p_R are Gaussians with diagonal covariance and we adopt the accuracy schedule $\alpha(t) = \sigma_1^{-2t}$. For a single continuous feature, it makes the KL divergence proportional to a squared error between the true value and the receiver mean. For discretized atom and bond features, we represent each category by k_c in the one-dimensional continuous space and use the expectation of these centers

Table 1 Performance comparison on QM9 and ZINC250k datasets. The best results are highlighted in bold and the second best are underlined. Null values (-) in criteria indicate that statistics are unavailable from the original paper.

Method	QM9 ($ N \leq 9$)				ZINC250k ($ N \leq 38$)				Sampling Steps
	Valid(%) $\uparrow$	Unique $\uparrow$	FCD $\downarrow$	NSPDK $\downarrow$	Valid(%) $\uparrow$	Unique $\uparrow$	FCD $\downarrow$	NSPDK $\downarrow$	
EDP-GNN	47.52	<u>99.25</u>	2.680	0.0046	82.97	99.79	16.737	0.0485	1000
GDSS	95.72	98.46	2.900	0.0033	97.01	99.64	14.656	0.0195	1000
DiGress	98.19	96.20	<u>0.095</u>	0.0003	94.99	99.97	3.482	0.0021	500
LGD-large	99.13	96.82	0.104	<u>0.0002</u>	-	-	-	-	1000
GruM	<u>99.69</u>	96.90	0.108	<u>0.0002</u>	98.65	99.97	2.257	0.0015	1000
DeFoG	99.30	96.30	0.120	-	99.22	<u>99.99</u>	<u>1.425</u>	0.0008	1000
SID	99.67	95.66	0.504	0.0001	99.50	99.84	2.010	0.0021	500
TopBF	99.74	99.27	0.093	<u>0.0002</u>	<u>99.37</u>	100.00	1.392	0.0008	200

Table 2 Comparison on mean absolute error (MAE) between unconditional generation and PEG guidance generation.

Target	$\mu \downarrow$	HOMO $\downarrow$	μ & HOMO $\downarrow$
Unconditional	1.69 ± 0.05	0.96 ± 0.01	1.73 ± 0.02
+PEG	$\lambda = 0.5$	0.92 ± 0.04	0.71 ± 0.02
	$\lambda = 1.0$	0.63 ± 0.04	0.51 ± 0.01
	$\lambda = 1.5$	1.55 ± 0.07	0.83 ± 0.03

under p_O . Here it yields the continuous-time losses for atoms and bonds (Appendix 2 for a detailed derivation):

$$\mathcal{L}_{\mathbf{X}} = -\ln \sigma_1 \mathbb{E}_{t \sim \mathcal{U}(0,1), p_F(\boldsymbol{\theta}|\mathbf{X};t)} \left[\frac{\|\mathbf{X} - \hat{\mathbf{k}}_{\mathbf{X}}(\boldsymbol{\theta}, t)\|_2^2}{\sigma_1^{2t}} \right], \quad (19)$$

where σ_1 is the target standard deviation of the input distribution at $t = 1$. An analogous loss $\mathcal{L}_{\mathbf{A}}$ is defined for bond features and the base objective is $\mathcal{L}_{\mathbf{G}} = \mathcal{L}_{\mathbf{X}} + \mathcal{L}_{\mathbf{A}}$.

To inject explicit topology awareness into training, we further minimize the expected Quasi-Wasserstein transport work defined in Sec. 4.5:

$$\mathbb{E}_{t \sim \mathcal{U}(0,1), \boldsymbol{\theta}_t \sim p_F(\boldsymbol{\theta}|\mathbf{G};t)} \left[\mathcal{J}_{\text{QW}}^{\mathbf{X}}(\boldsymbol{\theta}_t; \mathbf{G}) + \mathcal{J}_{\text{QW}}^{\mathbf{A}}(\boldsymbol{\theta}_t; \mathbf{G}) \right]. \quad (20)$$

This joint objective preserves the original Bayesian flow formulation while encouraging predictions that align with the ground-truth graph through low-cost geodesic transport.

4.7. Conditional Generation

TopBF supports conditional generation without retraining the unconditional flow model. Integrating property control via classifier-free (Ho and Salimans 2021) or regressor-free (Li et al. 2024) guidance often necessitates retraining models from scratch for each target property, which limits their flexibility across different tasks. Instead, we train a separate regressor and apply property-error gradient (PEG) guidance during sampling, which remains flexible for changing targets or composing multiple objectives.

Let $\mathcal{D}_{\text{prop}} = \{(G_i, \mathbf{z}_i)\}_{i=1}^M$ be a property-labeled dataset with property $\mathbf{z} \in \mathbb{R}^{d_z}$. We train a lightweight regressor R_{φ} on noisy graphs produced by the same sender corruption used in TopBF so that the regressor predicts the clean property from noisy inputs:

$$\hat{\mathbf{z}}_t = R_{\varphi}(\mathbf{X}_t, \mathbf{A}_t, t), \quad (21)$$

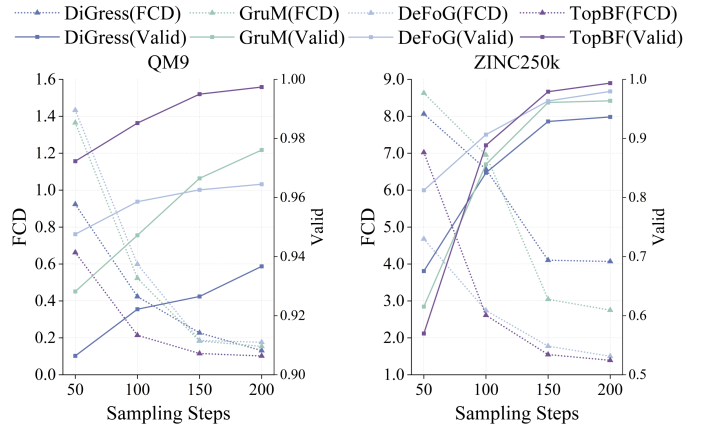


Figure 3 Sampling efficiency comparison between TopBF and baselines on QM9 and ZINC250k datasets.

$$\mathcal{L}_{\text{reg}} = \mathbb{E}_{(G, \mathbf{z}) \sim \mathcal{D}_{\text{prop}}} \mathbb{E}_{t \sim \mathcal{U}(0,1)} \mathbb{E}_{\mathbf{Y} \sim p_S(\cdot|\mathbf{G};\alpha(t))} [\|\hat{\mathbf{z}}_t - \mathbf{z}\|_2^2]. \quad (22)$$

Consequently, R_{φ} learns to estimate properties consistently along the BFN corruption path.

During sampling, we aim to obtain molecules with property close to specified target $\mathbf{z}^*$. Given the current expected class centers under $p_O(\cdot | \boldsymbol{\theta}_t)$, we compute $\hat{\mathbf{z}}_t = R_{\varphi}(\mathbf{G}_t, t)$ and define a property error:

$$\mathcal{L}_{\text{prop}}(\mathbf{G}_t; \mathbf{z}^*) = \|\hat{\mathbf{z}}_t - \mathbf{z}^*\|_2^2. \quad (23)$$

Before applying the Bayesian update, we take a small gradient step in parameter space

$$\tilde{\boldsymbol{\theta}}_t = \boldsymbol{\theta}_t - \lambda \nabla_{\boldsymbol{\theta}_t} \mathcal{L}_{\text{prop}}(\mathbf{G}_t; \mathbf{z}^*), \quad (24)$$

where $\lambda \geq 0$ controls the guidance strength and $\lambda = 0$ recovers the unconditional sampling.

5. Results

5.1. Baselines

We compare our model against several excellent baselines in MGG. EDP-GNN (Niu et al. 2020) models graphs with a permutation invariant approach. GDSS (Jo et al. 2022) is a score-based diffusion model. DiGress (Vignac et al. 2023) generates graphs using a discrete diffusion process. LGD (Zhou et al. 2024) provides a unified

Table 3 Ablation study on Quasi-Wasserstein transport coupling. The best results are highlighted in bold and the second best are underlined.

QW on		QM9 ($ N \leq 9$)				ZINC250k ($ N \leq 38$)			
Atoms	Bonds	Valid(%) $\uparrow$	Unique $\uparrow$	FCD $\downarrow$	NSPDK $\downarrow$	Valid(%) $\uparrow$	Unique $\uparrow$	FCD $\downarrow$	NSPDK $\downarrow$
x	x	98.63	97.85	0.143	0.0003	94.75	99.03	3.959	0.0031
✓	x	98.79	<u>99.13</u>	<u>0.102</u>	0.0003	95.61	100.00	1.347	0.0011
x	✓	99.81	99.05	0.107	0.0002	<u>99.33</u>	99.99	2.016	<u>0.0009</u>
✓	✓	<u>99.74</u>	99.27	0.093	0.0002	99.37	100.00	<u>1.392</u>	0.0008

framework for all-level graph generation and prediction via latent diffusion. GruM (Jo et al. 2024) models the generative process as a mixture of endpoint-conditioned diffusion. DeFoG (Qin et al. 2025) using a discrete flow matching framework that defines continuous-time probability paths over categorical attributes. SID (Boget 2025) assumes conditional independence between intermediate states to simplify discrete diffusion.

5.2. Molecular Graph Generation

Table 1 summarizes the main results on QM9 and ZINC250k. Overall, TopBF performs strongly on both datasets and achieves the best or tied-best results on most metrics while using only 200 sampling steps. This is noteworthy because the main competing diffusion and flow-based methods typically require 500 or 1000 steps. The results therefore suggest that TopBF improves sample quality without relying on long sampling trajectories.

On QM9, TopBF attains the best validity (99.74%), the best uniqueness (99.27), and the best FCD (0.093), while remaining tied for second on NSPDK (0.0002). Although the absolute improvement in FCD over the strongest baselines is not large, the comparison is already in a near-saturated regime on QM9, where further gains are difficult to obtain. In this setting, the simultaneous improvement in FCD and uniqueness is particularly meaningful, as it indicates that TopBF does not improve distribution matching by sacrificing diversity. Instead, it produces valid and diverse molecules whose overall distribution remains closer to the reference set.

On ZINC250k, which is more challenging because of larger graphs and a richer atom vocabulary, TopBF shows a favorable overall profile. It achieves perfect uniqueness at the reported precision, the best FCD (1.392), and the best-tied NSPDK (0.0008), while maintaining high validity (99.37%). Compared with DeFoG, TopBF further improves FCD from 1.425 to 1.392 with substantially fewer sampling steps. Compared with SID, TopBF is slightly lower in raw validity (99.37% versus 99.50%) but yields a markedly better FCD (1.392 versus 2.010), suggesting a better match to the global chemical distribution. Taken together, these results indicate that TopBF offers a balanced improvement in validity, diversity, and distributional fidelity across both easy and difficult benchmarks. Randomly selected generated molecules are provided in Appendix 7.

5.3. Conditional Generation

We next examine whether TopBF supports controllable generation on QM9. From the QM9 test split we randomly select 100 molecules, record their dipole moment μ and HOMO energy, and use each pair (μ^* , HOMO *) as a target. For every target we generate 10 molecules unconditionally and conditionally with PEG guidance, respectively. μ and HOMO are computed with Psi4 (Smith et al.

2020) from conformers built with RDKit ¹, and mean absolute error (MAE) is reported between the targets and the estimated properties for above three tasks. As shown in Table 2, PEG guidance consistently reduces the MAE relative to unconditional sampling for all three tasks, namely dipole moment μ , HOMO energy, and joint conditioning on both properties. The best performance is obtained at $\lambda = 1.0$, where the MAE decreases from 1.69 to 0.63 for μ , from 0.96 to 0.51 for HOMO, and from 1.73 to 0.77 for joint conditioning. These results show that TopBF can be steered effectively towards desired molecular properties without retraining the base generative model. We also observe a clear dependence on the guidance strength: $\lambda = 0.5$ improves control but remains conservative, whereas $\lambda = 1.5$ weakens performance relative to $\lambda = 1.0$, indicating that overly strong guidance may disturb the original sampling trajectory.

5.4. Sampling Efficiency and Effect

Figure 3 compares TopBF with representative baselines under different sampling budgets. TopBF reaches a strong performance within the first 100 steps and continues to improve with additional steps, although the gains become more gradual afterwards. Based on this trend, we use 200 steps as the default setting, which provides a favorable balance between generation quality and computational cost. Under this budget, TopBF remains competitive with, and in several cases superior to, methods that require considerably longer trajectories. This behaviour is consistent with the Bayesian flow formulation, where the model directly evolves distribution parameters and produces useful categorical confidence early in the sampling process.

5.5. Ablation on Quasi-Wasserstein Coupling

Table 3 reports the ablation study on the proposed QW transport coupling. Removing QW entirely leads to a clear degradation on both datasets. On QM9, validity drops from 99.74% to 98.63% and FCD worsens from 0.093 to 0.143. On ZINC250k, the effect is more pronounced, with validity decreasing from 99.37% to 94.75% and FCD deteriorating from 1.392 to 3.959. This indicates that the topology-aware transport objective contributes substantially beyond the base Bayesian flow loss.

The partial ablations further suggest that atom-level and bond-level coupling play different roles. Enabling QW on atoms mainly improves distributional fidelity, reflected by the strongest FCD on ZINC250k (1.347) and high uniqueness on both datasets. In contrast, enabling QW on bonds mainly benefits structural correctness, giving the best validity on QM9 (99.81%) and strong NSPDK on both datasets. When both components are used together, TopBF achieves the best overall balance across validity, uniqueness, FCD, and

¹ <https://www.rdkit.org>

NSPDK. This supports the design choice of jointly coupling node and edge variables in the transport objective.

6. Conclusion

In this paper, we introduce TopBF, a transport-coupled Bayesian Flow framework for molecular graph generation. TopBF operates in continuous parameter distributions and employs a CDF-based output channel to directly compute category probabilities, aligning the training objective with the discrete discretization used during sampling procedure, rather than fitting arbitrary continuous embeddings. To further inject topology awareness without breaking closed-form Bayesian updates, we formulate a Quasi-Wasserstein transport coupling under geodesic costs, encouraging topology-consistent alignment of atom and bond categorical mass on molecular graphs. Empirically, TopBF achieves state-of-the-art results on standard molecular graph generation benchmarks with efficient sampling of the minimum steps.

Supplementary data

Supplementary data are available as an attachment.

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